Hyperparameter Tuning and Cross Validation

★ Hyperparameter :-

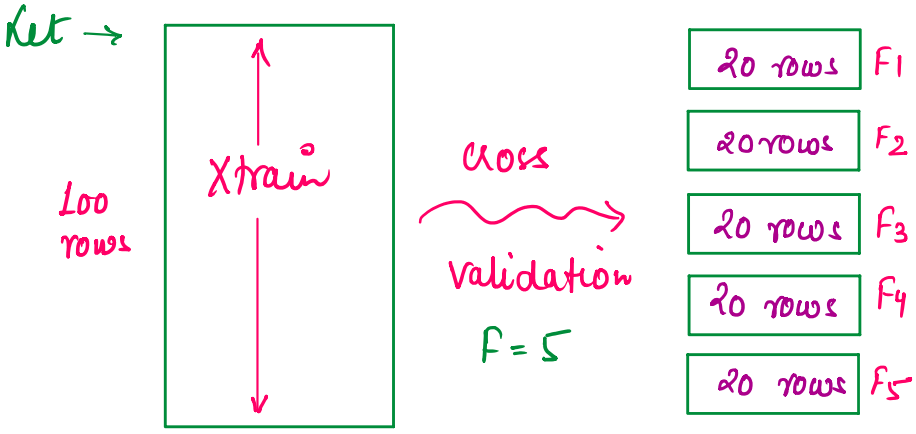
↳ An external value used before training a model in order to control its learning.

Hyperparameter Tuning :-

↳ Process of finding an optimal value to get the best performance of the model.

★ Cross Validation :-

↳ It is a technique used to evaluate how well a model generalizes to unseen data.



During model training,

	Training Set	Test Set
(1)	F_1 F_2 F_3 F_4	F_5
(2)	F_1 F_2 F_3 F_5	F_4
(3)	F_1 F_2 F_4 F_5	F_3
(4)	F_1 F_3 F_4 F_5	F_2

⑤	F_1 F_2 F_4 F_5	F_3
④	F_1 F_3 F_4 F_5	F_2
⑤	F_2 F_3 F_4 F_5	F_1

The most popular or widely used technique is K-fold cross validation

Before applying cross validation, we split the data into train & test sets

But, the dataset can also be divided into three parts :-

1. Training data
2. Validation data
3. Test data

* Hyperparameter Tuning Techniques :-

1. Grid Search CV
2. Randomised Search CV

- Grid Search CV :-

Let, KNN model n_neighbors, p & weights as hyperparameters

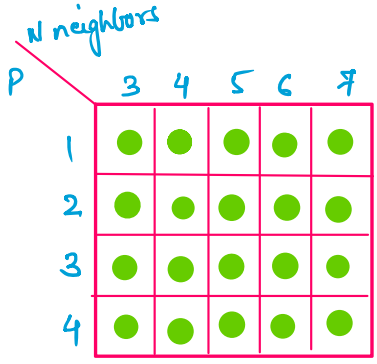
n_neighbors = [3, 4, 5, 6, 7]

p = [1, 2, 3, 4]

weights = ['uniform', 'distance']

Suppose, we take two hyperparameters

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Using grid search, we can try each & every combination in order to train the model & checks its performance.

So, whichever combination gives us best results, we finalise those as best hyperparameters

Drawback :-

↳ It is time consuming

↳ Not good for huge dataset

- Randomized Search CV :-

↳ Here we do not try for each & every combination.

↳ Here we take some random combinations.

